

Marcelo Andrés Oyarzo Catalán

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About me	A theoretical physicist and postdoctoral researcher at IGFA, working on gravity and related aspects of quantum field theory.	
Employment	2026-present	Postdoctoral Researcher, IGFAE Universidade de Santiago de Compostela
Education	PhD in Physics	2022–2025
	Double degree at <i>Universidad de Concepción, Chile & Politecnico di Torino, Italy</i> . Thesis: Black holes and Solitons in string theory Advisors: Andrés Anabalón, Julio Oliva, Mario Trigiante.	
	Master in Physics	2020–2022
	<i>Universidad de Concepción, Chile</i> Thesis: Black holes and solitons in supergravity and their exact scalar (quasi) normal modes Advisors: Julio Oliva, Fabrizio Canfora	
	Bachelor in physics	2016–2020
	<i>Universidad de Concepción, Chile</i>	
Grants	FONDECYT regular 1262452	co-investigador
	FONDECYT regular 1250133	co-investigador
	FONDECYT regular 11200025	personal técnico
	FONDECYT iniciación 11221063	personal técnico
Fellowships	PhD Fellowship (cotutelle)	Mar. 7, 2022 – Oct. 3, 2024
	<i>Universidad de Concepción, Chile / Politecnico di Torino, Italy</i> — Funded by Agencia Nacional de Investigación y Desarrollo (ANID).	
	Master Fellowship	Mar. 30, 2020 – Mar. 1, 2022
	<i>Universidad de Concepción, Chile</i> — Funded by Agencia Nacional de Investigación y Desarrollo (ANID).	
Peer Review	2025	“ <i>Annals of Physics</i> ”, Reviewer, Elsevier.

Teaching Assistant	2022	Classical Mechanics.
	2021	Statistical Physics.
	2020	Introduction to Particle Physics and Quantum Field Theory.
	2018	Differential and Integral Calculus.
	2017	Electromagnetism for Engineering Students.
Organization of Events	Jul. 2021	“ <i>School on Holography and Supergravity</i> ”, Online School.
	Jan. 2021	“ <i>School on Classical and Quantum Black Holes</i> ”, at Universidad de Concepción.
	2020-2021	“ <i>Gravity Group Seminar at UdeC</i> ”, Online seminars.

Publications

Preprints	Alejandro Alvarado, Andres Anabalón, Mariano Chernicoff, Julio Oliva, Marcelo Oyarzo, Gabriel Ortega, Jorge Urbina, “ <i>Logarithmic corrections to the entropy of near-extremal black holes in Einstein-Gauss-Bonnet</i> ”, [arXiv:2603.24939]
	Andrés Anabalón, Horatiu Nastase, Carlos Nunez, Marcelo Oyarzo, Ricardo Stuardo, “ <i>Moduli space of $N=4$ Super Yang-Mills from AdS/CFT</i> ”, [arXiv:2603.18141]
Published	Stefano Maurelli, Ruggero Noris, Marcelo Oyarzo, Mario Trigiante, “ <i>Warped-AdS3 Near-Horizon Geometries From TsT Transformations</i> ”, [arXiv:2512.01770] JHEP 03 (2026) 186
	Marcelo Oyarzo, Ricardo Stuardo, “ <i>(Iso)spin from Isospin in Top-Down Holography</i> ”, [arXiv:2512.02120] Accepted for publication in JHEP, 2025
	Pietro Fré, Federico Milanese, Marcelo Oyarzo, Matteo Santoro, Mario Trigiante, “ <i>Tessellation Groups, Harmonic Analysis on Non-compact Symmetric Spaces and the Heat Kernel in view of Cartan Convolutional Neural Networks</i> ”, [arXiv:2508.16015] Accepted for publication in Fortschritte der Physik, 2025
	Stefano Maurelli, Ruggero Noris, Marcelo Oyarzo, Henning Samtleben, Mario Trigiante, “ <i>Supersymmetric warped solutions from Type IIB orientifold reduction</i> ”, [arXiv:2504.16822] JHEP 08 (2025) 013, 2025
	Andrés Anabalón, Stefano Maurelli, Marcelo Oyarzo, Mario Trigiante, “ <i>The instability of low-temperature black holes in gauged $N = 8$ supergravity</i> ”, [arXiv:2411.09454] JHEP 03 (2025) 201, 2024
	Fabrizio Canfora, Nicolás Grandi, Marcelo Oyarzo, “ <i>Existence of a supersymmetric extension of the gauged nonlinear Σ model and the transition from integrability to chaos</i> ”, [arXiv:2411.04245] Phys.Rev.D 111 (2025) 11, 114026, 2024
	Jose Figueroa, Gaston Giribet, Anibal Neira-Gallegos, Julio Oliva, Marcelo Oyarzo, “ <i>Overflying nilpotent horizons</i> ”, [arXiv:2404.18378] Eur.Phys.J.C 84 (2024) 11, 1215, 2024
	Felipe Agurto-Sepúlveda, Julio Oliva, Marcelo Oyarzo, Dominik R. G. Schleicher, “ <i>Black hole shadows of α'-corrected black holes</i> ”, [arXiv:2404.12811] Phys.Rev.D 110 (2024) 2, 2024

Andrés Anabalón, Horatiu Nastase, Marcelo Oyarzo, “*Supersymmetric AdS solitons and the interconnection of different vacua of $N = 4$ Super Yang-Mills*”, [[arXiv:2402.18482](#)] JHEP 05 (2024) 217, 2024

José Barrientos, Adolfo Cisterna, Ivan Kolář, Keanu Müller, Marcelo Oyarzo, Konstantinos Pallikaris, “*Mixing “Magnetic” and “Electric” Ehlers–Harrison transformations: the electromagnetic swirling spacetime and novel type I backgrounds*”, [[arXiv:2401.02924](#)] Eur.Phys.J.C 84 (2024) 7, 724, 2024

Carlos Nunez, Marcelo Oyarzo, Ricardo Stuardo, “*Confinement and D5-branes*”, [[arXiv:2311.17998](#)] JHEP 03 (2024) 080, 2023

Fabrizio Canfora, Nicolas Grandi, Marcelo Oyarzo, Julio Oliva, “*Exact oscillations and chaos on a non-Abelian coil*”, [[arXiv:2309.04915](#)] Nucl.Phys.B 1004 (2024) 116553, 2023

Carlos Nunez, Marcelo Oyarzo, Ricardo Stuardo, “*Confinement in $(1 + 1)$ dimensions: a holographic perspective from I-branes*”, [[arXiv:2307.04783](#)] JHEP 09 (2023) 201, 2023

Carla Henríquez-Báez, Julio Oliva, Marcelo Oyarzo, Marcel Yáñez-Reyes, “ *R^2 corrections to the black string instability and the boosted black string*”, [[arXiv:2212.07296](#)] Phys.Rev.D 107 (2023) 4, 2022

Felipe Agurto-Sepúlveda, Mariano Chernicoff, Gaston Giribet, Julio Oliva, Marcelo Oyarzo, “*Slowly rotating and accelerating α' -corrected black holes in four and higher dimensions*”, [[arXiv:2207.13214](#)] Phys.Rev.D 107 (2023) 8, 2022

José Barrientos, Adolfo Cisterna, Cristóbal Corral, Marcelo Oyarzo, “*Gravitational instantons with conformally coupled scalar fields*”, [[arXiv:2202.13854](#)] JHEP 05 (2022) 110, 2022

Monserrat Aguayo, Ankai Hernández, José Mena, Julio Oliva, Marcelo Oyarzo, “*Exact scalar (quasi-)normal modes of black holes and solitons in gauged SUGRA*”, [[arXiv:2101.00438](#)] JHEP 07 (2022) 021, 2022

Fabrizio Canfora, Julio Oliva, Marcelo Oyarzo, “*New BPS solitons in $N = 4$ gauged supergravity and black holes in Einstein–Yang–Mills–dilaton theory*”, [[arXiv:2111.11915](#)] JHEP 02 (2022) 057, 2021

Nicolas Caceres, Jose Figueroa, Julio Oliva, Marcelo Oyarzo, Ricardo Stuardo, “*Quadratic gravity and conformally coupled scalar fields*”, [[arXiv:2001.01478](#)] JHEP 04 (2020) 157, 2020

Conference papers

Stefano Maurelli, Ruggero Noris, Marcelo Oyarzo, Mario Trigiante, “*Deforming $\text{AdS}_3 \times S^3 \times T^4$ in Type IIB Supergravity*”, [[arXiv:2604.26854](#)] , PoS CORFU2025 (2026) XXX, XXXX

Stefano Maurelli, Ruggero Noris, Marcelo Oyarzo, Mario Trigiante, “*On U-Folds and Their Construction*”, [[arXiv:2504.16822](#)] , PoS CORFU2024 (2025) 178, 2025

Jose Figueroa, Marcelo Oyarzo, “*Slowly rotating black holes modeled by Solv geometries*”, [[arXiv:2102.02328](#)] 22nd Chilean Physics Symposium 2020, 2021

Talks

- Dec. 16, 2025 “*Neural Networks and Rotating Black Holes in Higher-Curvature Gravities*”, Seminar, Universidad de Concepción, Chile
- Apr. 13, 2022 “*Hablemos del tiempo: de la concepción Newtoniana a la Relatividad General*”, Outreach Talk, Liceo Rodolfo Amando Philippi de Paillaco, Paillaco, Chile
- Apr. 13, 2025 “*Stability of Black Holes in the STU Model of $D=4$, $N=8$ Gauged Supergravity*”, Seminar, IGFAE, Universidade de Santiago de Compostela, Spain
- Mar. 20, 2025 “*Stability of Black Holes in the STU Model of $D=4$, $N=8$ Gauged Supergravity*”, Seminar, Università degli Studi di Torino, Torino, Italy
- Oct. 31, 2024 “*Black Holes in the STU Model of Gauged $D=4$ $N=8$ SUGRA*”, Seminar, Université libre de Bruxelles, Brussels, Belgium
- Oct. 17, 2024 “*Black Holes in the STU Model of Gauged $D=4$ $N=8$ SUGRA*”, Seminar, ICC Universitat de Barcelona, Barcelona, Spain
- Oct. 15, 2024 “*Black Holes in the STU Model of Gauged $D=4$ $N=8$ SUGRA*”, Seminar, Universidad de Oviedo, Oviedo, Spain
- Oct. 04, 2024 “*Novel solitonic background in type IIB supergravity*”, Seminar, Università degli Studi di Milano-Bicocca, Milan, Italy
- Sep. 23, 2024 “*Black Holes in the STU Model of Gauged $D=4$ $N=8$ SUGRA*”, Gong Show, V Siembra-HoLAGrav Young Frontiers Meeting, Online
- May 27, 2024 “*Supersymmetric AdS solitons and phase transitions in $N = 4$ SYM*”, Seminar, ICTP, Trieste, Italy
- Apr. 10, 2024 “*Supersymmetric AdS solitons in type IIB*”, Journal Club, Università di Torino, Torino, Italy
- Feb. 27, 2024 “*Supersymmetric AdS solitons and phase transitions in $N = 4$ SYM*”, Seminar, Swansea University, Swansea, Wales
- Dec. 18, 2023 “*A holographic perspective of a confining field theory in type IIB*”, Seminar, Politecnico di Torino, Torino, Italy
- Nov. 23, 2023 “*(1+1) confining QFT in type IIB string theory*”, Poster, CosmoConce conference, Universidad del Bio-Bio, Concepción, Chile
- Nov. 17, 2023 “*A holographic perspective of a confining field theory in type IIB*”, Seminar, Pontificia Universidad Católica, Santiago, Chile
- Nov. 16, 2023 “*A holographic perspective of a (1+1) confining field theory*”, Talk, Haifa University, Haifa, Israel
- Nov. 15, 2023 “*A holographic perspective of a (1+1) confining field theory*”, Seminar, Udec Gravity Group, Concepción, Chile
- Oct. 13, 2023 “*Periodic Solutions and Chaos in a Non-Abelian Coil*”, Seminar, Siembra-HoLAGrav Future of Physics, Online
- Oct. 04, 2023 “*A holographic perspective of a (1+1) confining field theory*”, Gong Show, IV Siembra-HoLAGrav Young Frontiers Meeting, Online
- Jun. 09, 2023 “*(1+1) confining QFT in type IIB string theory*”, Poster, Holography@25, ICTP SAIFR, Sao Paulo, Brazil

Jan. 18, 2023	“ <i>Confining and deconfining backgrounds in type IIB supergravity</i> ”, Poster, Nord-Grav@ICEN UNAP, Iquique, Chile
Oct. 16, 2022	“ <i>Perturbative higher curvature corrections of black holes and black strings</i> ”, Seminar, Centra, Universidade de Lisboa, Lisbon, Portugal
Oct. 14, 2022	“ <i>Perturbative higher curvature corrections of black holes and black strings</i> ”, Talk/Pre-Seminar, Centre de Physique Théorique, Paris, France
Oct. 14, 2022	“ <i>Perturbative higher curvature corrections of black holes and black strings</i> ”, Seminar, Université Libre de Bruxelles, Brussels, Belgium
Sep. 27, 2022	“ <i>Perturbative higher curvature corrections of black holes and black strings</i> ”, Talk, Swansea University, Swansea, Wales
Sep. 22, 2022	“ <i>Perturbative higher curvature corrections of black holes and black strings</i> ”, Seminar, Relativity Seminar at Institute of Mathematics of the Czech Academy of Sciences, Prague, Czech Republic
Sep. 22, 2022	“ <i>New black holes and solitons in gauged supergravity and their scalar (quasi-)normal modes</i> ”, Poster, Gravity@Prague 2022, Charles University, Prague, Czech Republic
Sep. 22, 2022	“ <i>Perturbative higher curvature corrections of black holes in string-inspired scenarios</i> ”, Talk, Cosmology and Particles 2022 UBB, Chillan, Chile
Apr. 25, 2022	“ <i>Gravitational instantons with conformally coupled scalar fields</i> ”, Seminar, Gravity group of Universidad de Concepción, Concepción, Chile
Apr. 25, 2022	“ <i>Black holes and solitons in supergravity</i> ”, Talk, 50 year of dimensional regularization, La Plata, Argentina
Apr. 14, 2022	“ <i>Gravitational instantons with conformally coupled scalar fields</i> ”, Seminar, Terapia de grupo, Universidad de Buenos Aires, Buenos Aires, Argentina
Apr. 13, 2022	“ <i>¿Qué hay en el interior de los agujeros negros?</i> ”, Outreach Talk, Liceo Rodolfo Amando Philippi de Paillaco, Paillaco, Chile
Apr. 06, 2022	“ <i>Black holes and solitons in supergravity and their exact scalar (quasi-)normal modes</i> ”, Talk, Kickoff Meeting Holography CL, Valparaiso, Chile
Oct. 07, 2021	“ <i>Non-Abelian black holes in Einstein-Yang-Mills-Dilaton and their embedding in supergravity</i> ”, Seminar, Universidad de Concepción, Concepción, Chile
Sep. 10, 2020	“ <i>The phase space of constrained systems: How to proceed when the $\dot{q}$'s cannot be solved in terms of the p's</i> ”, Seminar, Weekly seminars at gravity group of Universidad de Concepción, Concepción, Chile
Sep. 06, 2020	“ <i>New topological Lifshitz black holes in Einstein-Yang-Mills-Dilaton with rich thermodynamics properties</i> ”, Gong Show, II Siembra-HoLAGrav Young Frontiers Meeting, Online
Mar. 08, 2020	“ <i>Quadratic gravity and conformally coupled scalar fields</i> ”, Talk, Escuela de gravitación y Relatividad General, Valdivia, Chile

Participation in Schools and Conferences

Sep. 23–25, 2024	“ <i>V Siembra-HoLAGrav Young Frontiers Meeting</i> ”, Online meeting organized by Siembra-HoLAGrav
Apr. 22 – Jul. 4, 2024	“ <i>Supersymmetry II and AdS/CFT</i> ”, Course at SISSA
Nov. 22–24, 2023	“ <i>CosmoConce</i> ”, Conference, Universidad del Bio-Bio, Concepción, Chile
Jan. 2024	“ <i>Workshop on Theoretical High Energy Physics 2023</i> ”, Conference, UCSC, Concepción, Chile
Jan. 2024	“ <i>NordGrav@ICEN</i> ”, Conference and School, UNAP, Iquique, Chile
Oct. 3 – Nov. 28, 2022	“ <i>Brussels-Paris-Geneve Doctoral Solvay School</i> ”, School
Sep. 19–23, 2022	“ <i>Gravity@Prague</i> ”, School, Prague, Czech Republic
Apr. 25–26, 2022	“ <i>50 YEARS OF DIMENSIONAL REGULARIZATION</i> ”, Conference, La Plata, Argentina
Apr. 4–6, 2022	“ <i>Kickoff Meeting Holography CL</i> ”, Conference, Valparaiso, Chile
Sep. 2021	“ <i>II Siembra-HoLAGrav Young Frontiers Meeting</i> ”, Conference
Jul. 2021	“ <i>School of Holography and Supergravity</i> ”, School, Universidad de Concepción, Chile
Jan. 2021	“ <i>School on Classical and Quantum Black Holes</i> ”, School, Universidad de Concepción, Chile
Nov. 2020	“ <i>II School of Holography and Entanglement Entropy</i> ”, Siembra-HoLAGrav
Mar. 2020	“ <i>Escuela de gravitación y Relatividad General</i> ”, School, CECs
Jul. 2019	“ <i>Topics on Graviticulas</i> ”, Winter School HEP-PUC